

# Reading Hawking's Presence: An Interview with a Self-Effacing Man

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In *Mythologies*, Roland Barthes dissected, in two short pages, one of our mythical objects: the brain of Einstein. "Paradoxically," he said,

the greatest intelligence of all provides an image of the most up-to-date machine, the man who is too powerful is removed from psychology, and introduced into a world of robots; as is well known, the supermen of science-fiction always have something reified about them. So has Einstein: he is commonly signified by his brain, which is like an object for anthologies, a true museum exhibit. Perhaps because of his mathematical specialization, superman is here divested of every magical character; no diffuse power in him, no mystery other than mechanical: he is a superior, a prodigious organ, but a real, even a physiological one. . . . The mythology of Einstein shows him as a genius so lacking in magic that one speaks about his thought as of a functional labour analogous to the mechanical making of sausages, the grinding of corn or the crushing of ore: he used to produce thought, continuously, as a mill makes flour, and death was above all, for him, the cessation of a localized function: '*the most powerful brain of all has stopped thinking*'. What this machine of genius was supposed to produce was equations.<sup>1</sup>

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1. Roland Barthes, *Mythologies*, trans. Annette Lavers (1957; New York, 1972), pp. 68–69.

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In 2003 the brain machine is no longer a myth. There exists a scientist whose neuro-motor functions, such as speech, writing, and movement, have been partly made possible through the mediation of a machine: a computer. People say—as they did formerly of Einstein—that he is a genius. His name is Stephen Hawking.

Forty years ago Hawking developed amyotrophic lateral sclerosis, more commonly known as Lou Gehrig's disease. He is now totally paralyzed and confined to a wheelchair. In 1985 he definitively lost his ability to speak because of complications arising from pneumonia that required him to have a tracheotomy. Thanks to a communication program called Living Center (designed and given to him by Walt Woltoz of Words + , Inc.) and a speech synthesizer (designed and given to him by Speech Plus, Inc.), Hawking can today communicate, write, and read. He is able to operate this entire system on his own by means of a simple switch (the commutator) that someone places in his hand and around which he curls his fingers. The device consists of a highly sensitive on/off switch that operates a cursor. Since the cursor automatically moves up and down, Hawking can select words by pressing on the switch. When he has built up what he wants to say, he can send it to the speech synthesizer. The synthesizer and a small IBM-compatible personal computer (donated by David Mason of Cambridge Adaptive Communication) have both been mounted on his wheelchair. They run from batteries under the chair, although an internal backup battery will keep the computer running for up to an hour if necessary. The computer screen is mounted on the arm of the wheelchair where he can see it.<sup>2</sup> Hawking can either speak what he has written or save it to disk. He can then print it out or call it back and speak it sentence by sentence. As he has explained, "Using this system, I have written a book, and dozens of scientific papers. I have also given many scientific and popular talks."<sup>3</sup>

The case of Hawking is interesting insofar as it seems to reconcile two contrasting research traditions. On the one hand, he appears to be the very incarnation of Cartesian dualism, with his genius constructed in the pop-

2. See Stephen Hawking, "Disability: The Computer," <http://www.hawking.org.uk/text/disable/computer.html>

3. Hawking, "Disability: My Experience with ALS," <http://www.hawking.org.uk/text/disable/disable.html>

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ular media around the mind/body dichotomy. Thus, because of his handicap he is no longer distracted by the daily and worldly occupations shared by the rest of humanity; he can therefore devote himself entirely to thought.<sup>4</sup> According to Barthes, the mythology of Einstein is that his thoughts are materialized as a brain working like a machine; in the case of the wheelchair-bound Hawking, the opposite is true, for here mythology operates to entirely disincorporate his genius. In this sense, Hawking embodies the mythical figure of a genius capable of grasping the ultimate laws of the universe with nothing but the strength of his reasoning.

On the other hand, in trying to describe and understand more fully his way of working, I discovered that his "genius" validates one of the principal results obtained by sociologists and historians of science, namely, the profoundly located, material, and collective nature of the production of knowledge as opposed to the view that it is rationalist, individualist, and disincorporated. In other words, in order to understand the functioning of this genius's mind, one has to understand the work of delegation, multiplication, and redistribution of competences to people and machines without which he would be unable to think. I refer here specifically to the mobilization of the competences of his computer; his students, spread across very different research fields (which enable him to integrate diverse information and different facets of a problem); and his secretary, who sorts and arranges data according to his interests and to what his brain can process.<sup>5</sup> Following from this, one might conclude, turning the problem upside-down, that all scientists are *in some sense* handicapped insofar as they are unable to think without being attached to a set of instruments, machines, and laboratories through which their intelligence is distributed. This is why Hawking, contrary to the image generally presented of him, does not have a mind free from all worldly constraints; it is precisely because he suffers from a real handicap that I have discovered that he is more incorporated than any other theoretician. Indeed, since he is unable to make his own body function, all the motor and cognitive operations normally unseen by the ethnographer because incorporated in a single body (as in the case of a healthy physicist) are delegated and incorporated in other bodies.<sup>6</sup>

Thus in a certain sense, Hawking's handicap becomes a window both on

4. For an examination of the relationship between asceticism and truth, see Steven Shapin and Christopher Lawrence, introduction to *Science Incarnate: Historical Embodiments of Natural Knowledge*, ed. Lawrence and Shapin (Chicago, 1998), pp. 1–19.

5. She will, for example, show him invitations to conferences but not letters from his admirers, and takes charge of the professional as much as the private side of his person.

6. See Hélène Mialet, "Do Angels Have Bodies? Two Stories about Subjectivity in Science: The Cases of William X and Mr. H," *Social Studies of Science* 29 (Aug. 1999): 551–82.

the *representation* of how science and scientists work and on the actual practice of science. On the one hand, it shows us (through the analysis of the popular media) what a perfect scientist is or should be. On the other hand (through ethnographic analysis), it shows us *not* that Hawking is different from other scientists *because* he has a handicap, but rather that *because* he has this handicap he gives us the possibility of seeing how theoretical scientists (supposedly using only their minds) work.<sup>7</sup>

The notion of seeing the normally unseen is, however, extremely ambiguous precisely because when I began to think along these lines, Hawking was unapproachable.<sup>8</sup> Based on articles and interviews, my reconstruction of the network of humans and machines surrounding him was as much of a reconstruction as any other article. Moreover, I was left with the strange feeling that the more I worked with written representations and images of Hawking, the more I had a stable image of who he was—a man with a relatively well-defined ego.<sup>9</sup> Yet the closer I came to the scientist's body, the greater access I had to what I call his distributed or extended body (his secretary, his computer, and his students), but not to him, because he refused all my attempts to meet with him. Finally, when I read accounts of people who had met him, they typically reported that they were unable to tell whether they were dealing with a Hawking who was suffering, bored, or thinking. In short, the closer I was to the single body, the more I discovered a multiplicity of Hawkings, while the more I turned my focus to the multiplicity of constructions and representations of Hawking in the media, the more I could see a stable ego appear. My first investigation concerning Hawking's identity thus ended with the perhaps inevitable question: Where are you, then, Professor Hawking?

Eventually I did meet Professor Hawking. And this is still the question I am trying to answer.<sup>10</sup>

In this paper, I will recount the interview I had with him. But rather than focusing only on the content of the interview, I also want to present an ethnographic study of the interview itself. In this account, I will focus on the computer, on the intelligent system that mediates Hawking's relation-

7. This is one of the avenues that I am currently exploring in a book that I am writing on Hawking.

8. See Mialet, "Do Angels Have Bodies?"

9. Over and over again one finds the same jokes about his voice giving him an American accent, the role of his ex-wife in his survival, or his birth 300 years after the death of Galileo, all of which, I argued, were partly the result of the standardization of journalistic accounts and partly because they were stored in his computer.

10. For a description of a similar feeling created in the presence of Hawking, see Rosanne Allucquère Stone, "Split Subjects, Not Atoms; or, How I Fell in Love with My Prosthesis," *Configurations* 2 (Winter 1994): 173–90.

ships with others (including myself) as well as with himself. In doing so, I will try to explain the functioning of his extended body. We will see that Hawking does not think like a machine, as does Barthes's Einstein, but neither can he act *without* a machine or without his assistants who work day and night to maintain and stabilize his identity as Stephen Hawking the man and Stephen Hawking the Genius Physicist.

This paper addresses a number of different questions:

1. The notion of *mise en présence*, the *to be there* so dear to anthropologists. But what does it mean in this particular context? Do we learn more about a person (in this case Hawking) when we are in his or her presence?<sup>11</sup>

2. Behind this methodological question there is another: what difference does it make when one deals with texts as opposed to a person?<sup>12</sup> With Hawking this question has a different resonance, for in his presence, I will argue, one is dealing with a form of transcript. I will show, then, that far from escaping the world of writing and representations to find ourselves in contact with a person, we instead are confronted with an illegible body and a text being written about Hawking, Lucasian Professor of Mathematics—a text that will be stored in a part of his computer, waiting to be added to other narratives about Hawking.

3. What kind of body is Hawking's body? Where is his body? I will show how Hawking's competences are exteriorized and incorporated in the human- and machine-based network around him, thus giving him the possibility of maintaining his identity as Stephen Hawking the man and Stephen Hawking the Genius Physicist, while at the same time deleting the presence of his flesh-and-blood body. Following from this, I will also try to explain how and why the interview itself became entangled in these processes of embodiment and disembodiment.

4. Finally, what can we say about the role of machines? Though machines or instruments are often represented as transparent extensions of the body, in this case they play a more active role. I will describe how the machines and instruments around Hawking disrupt as much as they make possible the interaction while making it disappear at the same time. In this sense,

11. In a way, what I try to question in this context is a certain ideology claiming the primacy of the field and the immediacy of experience that accompanies it, the idea that the *mise en présence* with the other would give the analyst better access to the true nature of things. I also engage with the idea, already put into question by Erving Goffman, that it is possible to get rid of the different layers that constitute a person so as to reach "the true self."

12. This question could be linked to certain reflections about the relationship between historiography and anthropology. See Paul Ricoeur, "What Is a Text? Explanation and Understanding" and "The Model of the Text: Meaningful Action Considered as a Text," *Hermeneutics and the Human Sciences: Essays on Language, Action, and Interpretation*, trans. and ed. John B. Thompson (Cambridge, 1981).

we will see how the relation of power between the “anthropologist” and the “native” shifts. Indeed, the establishment of the text is central to what ethnographers do both in the field and thereafter.<sup>13</sup> In this case, however, the “native” writes, registers, and edits the interview, leaving the ethnographer powerless and destabilized.<sup>14</sup>

### First Contact

25 June 1998. The man I have been trying to meet for over two years has agreed to meet with me today, one week before I am to leave England. I have found, it seems, the magic word, the key to pass through the gates guarding the Lucasian Professor of Mathematics. Each previous attempt failed. When I first wrote to Hawking (then to his personal assistant) as a philosopher of science excited by his discoveries, and then again as a sociologist keeping her ears open to the media events surrounding each of his moves, the answer was always the same: No. No, because the professor is too busy. No, because the professor thinks that it's not appropriate. The magic word—the open sesame—was *Lucasian Professor of Mathematics*. After being commissioned to write an article in a book on the history of the Lucasian professorship, I asked again for an appointment with Hawking.<sup>15</sup> One day later, an email was waiting for me saying, “Professor Hawking would like very much to meet you to discuss the chapter you are going to write in the said book. Could you, please, contact Pr. Hawking's Personal Assistant at the address below . . . Thomas Auld, graduate assistant to Professor Hawking.”

25 June 1998. 4:15 p.m. The waiting room of the department of physics and applied mathematics of Cambridge is swarming with students. It is tea time. Thomas Auld invites me to go into his office: a small office, almost empty, with a big table overlaid with two big black boxes on the left side, a laptop and a desktop computer on the right. A drawing of a wheelchair attached to a computer hangs on the wall. Auld explains to me that a tour has been organized for me with Stuart Rankin, the department computer officer, and Paul Shellard (a former student of Hawking who now collaborates with him and takes care of the administrative part of the department).

13. For a critique of the ideology that claims that representations produced by anthropologists—like keeping good field notes, making accurate maps, writing up results—are transparent, see James Clifford, “Introduction: Partial Truths,” in *Writing Culture: The Poetics and Politics of Ethnography*, ed. Clifford and George E. Marcus (Berkeley, 1986), pp. 1–26.

14. See, for example, Isabelle Stengers, who proposes that we give the subjects of the human sciences the power of putting into question the tools of the analyst; see Isabelle Stengers, *Pour en finir avec la tolérance* (Paris, 1997).

15. See Mialet, “Is the End in Sight for the Lucasian Chair? Stephen Hawking as Millennium Professor,” in *From Newton to Hawking: A History of Cambridge University's Lucasian Professors of Mathematics*, ed. Kevin Knox and Richard Noakes (Cambridge, 2003).

I just came to see Hawking and Auld; I didn't expect this tour. I was also surprised when Auld told me that if I wanted I could interview Hawking's nurse, Christine, as well. The technicians, apparently, are not invisible after all.<sup>16</sup>

Thus I have met, one after the other, Auld, Rankin, and Shellard. Hours have passed and I still haven't talked to Hawking, though I have seen him as my tour passed repeatedly by the open door of his office. A bit impatient at this point, I ask Shellard when I will be able to meet Hawking and if he thinks it might be possible to do so now. He gets up quickly and apologizes. "Of course! It's more important to interview Stephen Hawking than me." I wait in his office. I'm wondering why Hawking has agreed to meet me. His personal assistant confirmed that it was quite rare for him to accept interviews. Did he hear about my study? Does he want me to stop investigating him? How am I going to behave before this person I have tried to demystify but who nevertheless is still very intimidating?<sup>17</sup> How are we going to communicate? Is he going to use his artificial voice? Shellard brings an end to all these thoughts: "Stephen Hawking is ready to meet you." I follow Shellard into the office. Hawking, sitting in his wheelchair, is half hidden behind a large computer placed on the desk in front of him. I think about what Gene Stone said: "Almost everyone's first contact with him is metal."<sup>18</sup> For me, though, I'm struck and moved rather by his fragility. I greet Hawking. Shellard, seeing that I don't know where to sit, asks Stephen, then tells me to sit near Hawking. I ask if he would mind if I recorded our interview. Shellard seems to think that it's not a problem. With a smile on his face, he leaves the room. I can hear behind me as the nurse comes and goes—washing, straightening papers, and so forth. I have decided to focus my interview on his role as Lucasian Professor and not ask him any personal questions related to my interest in his practical way of working. I switch on the tape recorder and open the conversation:

H.M.: Thank you very much for accepting this interview, Professor, I know that you are very busy. Thomas Auld explained to you why I wanted to talk to you.

I stop. I'm not sure that he is ready. He seems to write something but I don't know where. After a few seconds of silence, I start again.

H.M.: Thank you very . . .

16. See Shapin, "The Invisible Technician," *American Scientist* 77 (Nov.–Dec. 1989): 554–63.

17. It would probably be interesting to analyze the relation between mystification and fear.

18. Arthur Lubow, "Heart and Mind," *Vanity Fair* (June 1992): 44–53. Stone was responsible for organizing the transcripts from the film interviews into *Stephen Hawking's "A Brief History of Time": A Reader's Companion*, ed. Hawking and Gene Stone (New York, 1992).

His answer to my first question stops me. A voice speaks from somewhere:

S.H.: I CAN GIVE YOU A PRINTOUT OF WHAT I SAY.<sup>19</sup>

H.M.: Thank you. It would be very helpful.

The tape recorder will nevertheless be on during the first part of the interview; I will be able to hear just the click-click of his commutator (the instrument he uses to spell out words) and my own questions.<sup>20</sup>

H.M.: So I told you I've been commissioned to write an article in a book on the history of the Lucasian Professor of Mathematics, so my chapter will be on you, on Professor Hawking; the idea of the book will be to examine the role of the Professor inside his cultural, intellectual, technological context. And so my first question is: Why is it so important for you to be the Lucasian Professor of Mathematics?

He begins writing. I realize that I force my voice to speak louder (strangely, I have the feeling that because he can't talk, he also can't hear). A sentence is written on the small computer attached to the arm of the chair (I'm not sure whether it was written on the big screen in front of him because I didn't see it immediately). It is accompanied by Hawking's synthetic voice. I will notice only later when I listen to the tape that his synthetic voice repeated the sentence too; at the time, for some reason, I didn't hear it.

S.H.: WHY YOU?

H.M.: Why you? Ah! Why me?

I'm surprised; I was expecting the answer to my question and not a question from him.

H.M.: Because I'm a philosopher of science, I'm a sociologist of science, I work at the Maison Française of Oxford, I'm running a program in History of Science, and I'm extremely interested in creativity and discovery. So I'm writing a book . . . I studied an inventor in France and I was interested in your discoveries because you are considered to be someone very creative, and so I was told it would be appropriate if I could write this chapter.

As I speak he is still working at answering my question. I can watch the answer slowly appearing on the screen:

19. I will use capital letters to indicate when the voice speaks.

20. The commutator amplifies and makes audible the curling of his fingers.



I didn't apply to be Lucasian professor. I was already an ad hominen professor at Cambridge so it wouldn't have raised my salary. I hoped the position would have been used to bring in someone good from outside Cambridge like atiyah. I was rather disappointed when George batchelor who was then head of DAMTP told me the electors had chosen me. I urged them to think again and get someone like atiyah. But Batchelor was very much against the idea of giving the Lucasian professor to Atiyah whom he regarded as a pure mathematician thinly disguised as a physicist. It might have meant that the Lucasian professor was lost from applied mathematics to pure. In the end I agreed to accept if I was given certain support. I must say I like the feeling that I hold the same job as newton and Dirac. When I joined DAMTP in 1962, it contained only two professors, the Lucasian professor, Paul Dirac, and the Plumian professor, Fred Hoyle. Shortly after that, there was a contest between Hoyle and Batchelor over who should be head of department. Hoyle lost and in disgust transferred his chair to the faculty of physics and chemistry. So for a time the Lucasian professor was the only professor in DAMTP. But now we have about five other established chairs and about six ad hominum professors, the Lucasian professor ship is not so important. It is a bit like the title, astronomer royal, that my colleague and fellow research student, Martin Rees has. It is nice to feel that one holds the same position as newton and Dirac, but the real challenge is to do work that is even a small fraction as significant. I think I was appointed as a stop gap to fill the chair as someone whose work would not disgrace the standards expected of the Lucasian chair but I think they thought I wouldn't live very long and then they choose again by which time there might be a more suitable candidate. Well I'm sorry to disappoint the electors. I have been Lucasian professor for 19 years and I have every intention of surviving another 11 to the retiring age. Even so I won't match Dirac who was Lucasian professor for 37 years or stokes who was for 54. I hope they get someone good after me but now that there are so many other professors, it won't have quite the same draw. I'm also proud to have babbage as one of my predecessors, even if he wasn't Lucasian professor for very long. I do a certain amount of fund raising for Cambridge university in places like silicon valley, and is a good line to throw in that the father of computers was one of my forebears.

FIGURE 1. Hawking's responses to my questions as printed out.

S.H.: I didn't apply to be Lucasian Professor. I was already an ad hominen professor at Cambridge so it wouldn't have raised my salary.<sup>21</sup>

The nurse asks me if I would like some tea. I say: "No, no thank you." And continue reading the answer:

S.H.: I hoped the position would have been used to bring in someone good from outside Cambridge like atiyah. I was rather disappointed when George batchelor who was then head of DAMTP told me the electors had chosen me. I urged them to think again and get someone like atiyah. But Batchelor was very much against the idea of giving the Lucasian professor to Atiyah whom he regarded as a pure mathe-

21. Hawking's responses to my questions have been transcribed exactly as they were printed out; I have kept his spelling without making any changes or corrections. See fig. 1 for a facsimile of the printout.

matician thinly disguised as a physicist. It might have meant that the Lucasian professor was lost from applied mathematics to pure. In the end I agreed to accept if I was given certain support. I must say I like the feeling that I hold the same job as Newton and Dirac.

For more than two hours, questions and answers follow, one after another. Five exactly. Five of the ten possible questions prepared in advance with the editor of the book. Five answers slowly written on the computer. At the end I will leave with Hawking's answers printed out, with no trace of the small talk framing the interview (like Hawking's question "Why you?" at the beginning and his assertion, "My answers were hard hitting and frank—I don't want you to water them down."). No trace of a whole range of interactions that occurred—through or thanks to the computer—which I will recall and describe here.

25 June 1998. 5:30 p.m. The room is now totally quiet. I'm alone with Hawking. The nurse sits at the entrance of the room. She says hello to someone who just walked by the door, talks quickly with Auld, then concentrates on her knitting needle. Both Hawking and I sit in silence looking at the computer. I have just asked him if he thought that the intellectual authority of the Lucasian Professors of Mathematics had changed. He writes. I wait, following the cursor on the screen.

S.H.: It is nice to feel that one holds the same position as Newton and Dirac, but . . . [suction]

Suddenly, the click-click of the commutator stops. In the middle of writing his answer this word crops up in his text, and the voice says "SUCTION." The nurse immediately arrives, takes a Kleenex, and wipes his mouth. Then she opens a small bag, takes a little pipe, puts it in Hawking's neck, and activates a machine. His body shakes. I think that I should perhaps leave the room, but Hawking, seemingly oblivious to the nurse, continues with his writing.

S.H.: . . . but the real challenge is to do work that is even a small fraction as significant.

His body has been taken care of and he can carry on with what he was doing: being the Lucasian Professor of Mathematics writing about the Lucasian Professor of Mathematics.

I then ask him a new question: "How do you imagine the future of the Lucasian Professor of Mathematics?" Hawking starts writing.

S.H.: I think I was appointed as a stop gap to fill the chair as someone whose work would not disgrace the standards expected of the Lucasian chair but I think they thought I wouldn't live very long and then

they choose again by which time there might be a more suitable candidate. Well I'm sorry to disappoint the electors. I have been Lucasian professor for 19 years and I have every intention of surviving another 11 to the retiring age. Even so I won't match Dirac who was Lucasian professor for 37 years or Stokes who was for 54. [legs]

Again, a word appears under the sentence he has just written. The voice says: "LEGS." The nurse arrives, delicately takes his leg, and moves it slowly. Then *chair-in* appears on the computer screen (though the voice stays silent). The nurse asks him if it concerns the batteries. She looks at the little computer resting on the arm of the wheelchair, then turns the wheelchair around and begins to manipulate a big black box under it marked MOZART. The voice says "NO," then "GET TOM." The nurse goes out and comes back with Tom. She explains to him that Stephen must have a problem; she thought that it was the battery. Tom looks quickly at the computer then inspects something under Hawking's chair. He leaves the room and comes back with a big black box with *Mahler* written on it. He replaces the black box settled under Hawking's chair with the new one. Hawking can now start writing again and finishes his response.

S.H.: I hope they get someone good after me but now that there are so many other professors, it won't have quite the same draw.

I ask the professor if he feels too tired or if I can ask him another question. The voice says "YES." I ask him my question: "You talk a lot about Newton and Dirac. What do you think about Babbage?"

The thumb starts caressing the commutator in his hand again; the words accumulate one after another on the screen. After fifteen minutes, the cursor stops moving. I never know if he has finished or not. I can read the answer:

S.H.: I'm also proud to have Babbage as one of my predecessors, even if he wasn't Lucasian professor for very long. I do a certain amount of fund raising for Cambridge university in places like silicon valley, and is a good line to throw in that the father of computers was one of my forebears.

It's almost 7:30 p.m., so I ask him if he wants me to leave. He writes, "Yes, I have to go." Then, "My answers were hard hitting and frank—I don't want you to water them down." I ask him if he wants me to show him the article before its publication. The voice says "YES." He writes, "When will the paper be finished?" I say, "Not before the end of next year." He smiles. (I interpret this smile as saying "you are as slow as I am.") I say, "Goodbye and thank you." He writes on the computer, "Can you get the nurse." He sees that I read the message and the voice is not activated. I go to get the nurse,

but she's not there. I try not to get panicky and decide to get Tom in his office. I open his door; he's making a phone call; I tell him that Hawking wants his nurse, but I can't find her. He immediately ends his call and goes to see Hawking. I follow him. When we arrive in the room the nurse is there. "So you were looking for me!" she says. She shows him some pictures and he smiles. I seize the opportunity to look at the copy of the famous bet he made with Kip Thorne about one of his hypotheses, which is hanging on the wall.<sup>22</sup> The nurse notices that I'm still here and asks me if I want something. I say that I wanted to say goodbye and thank the professor. I leave and go to see Tom. He leaves me briefly to pick up the printed answers to my interview. When he comes back, I mention that Hawking agreed to answer some of my questions by email. Tom adds that if I need anything, I should write to him and he will relay it to Hawking or organize interviews with other people I want to meet. It is time to leave. The cars I saw through the windows when I arrived three hours earlier have disappeared. Instead, I see our reflections in the window panes. The sun has set. I leave Tom and walk through the now silent and empty waiting room. I close the door of the physics and applied mathematics department, leaving behind me the well-protected building that houses the Lucasian Professor of Mathematics. The place is once again shrouded in mystery in the twilight of an English summer evening. That was 25 June 1998 at 8:00 p.m.

### *Mise en Présence: Reading a Body and/or a Text*

I would now like to explore the key role of the computer as the physical—and not metaphorical—extension of Hawking's body in the scene described above, especially with regard to the place it occupies. I am not face to face with Hawking, but like him, seated facing his computer. Or, rather, my body is facing the computer whereas my face is turned towards Hawking when I ask him questions (as I would in any interaction), for it is him and not the computer I am addressing. My head turns slightly, in line with my body, to read his answers displayed on the screen, that is to say, to read what I interpret as being his wishes, will, and thoughts. As for Hawking, he remains seated, an extension of the computer to which he is connected by his finger on the commutator. At times he tries, almost imperceptibly, to move his head, which is already leaning slightly towards me, to give a very faint smile as I talk to him. Thus, it is through this constant coming and going between a silent human body, capable of "talking" with its face (that is to say, of creating a presence, a contact) and a more or less silent mechanical body—<sup>23</sup> depending on whether or not Hawking decides, after having made his discourse visible,

22. See Kip S. Thorne, *Black Holes and Time Warps: Einstein's Outrageous Legacy* (New York, 1994), pp. 481–82.

23. Some machines are very noisy.

to make the machine “talk”—that I see the first link between these two bodies emerge. Without the computer I would understand nothing of what Hawking wants or wishes to express. Without Hawking, the computer would remain opaque and voiceless and my presence would be incongruous.<sup>24</sup> Yet because of this triangulation (Hawking, the computer, and myself) the reading of the interaction, that is, of what is wanted, said, and expected by the different actors, is totally muddled. In the first minutes I don’t know whether I should look at Hawking, the computer in front of him, or the one attached to his wheelchair. I can’t hear the voice talking to me; I don’t know whether or not Hawking has finished writing, whether I can or should laugh, wait, or carry on asking questions, stay or leave.

The practices of attuning to a conversation that we take for granted when we talk to someone endowed with similar competences are disrupted and put together again in different ways. Hawking’s unique conditions, the distinctive modalities with which he communicates and my unfamiliarity with them, thus tells us something about what is implied in a “normal” interaction or conversation.<sup>25</sup> We find ourselves confronted with a paradox. On the one hand, we have shown that Hawking, because of his handicap, makes visible that which is normally hidden in the scientist’s practice and self-presentation, for example, the role of his assistants and the machines to which he is attached.<sup>26</sup> On the other hand, his handicap also makes visible, because they are absent or because they are modified by his *mise en présence*, a set of elements that are visible (although unnoticed) and constituent of all social interaction, such as body language, intonation, the passage of time, and spatial relations.<sup>27</sup> Let us start with time.

24. Except if we communicated by email. See Harold Garfinkel, *Studies in Ethnomethodology* (Cambridge, 1967).

25. Like Garfinkel’s famous breaching experiment that consists in disturbing routines, Hawking’s disruption reveals something about the necessary requirements of fluent social interactions/conversation and/or about the disruption itself. It also tells us that who Hawking is reflects differences in the microsociology of face-to-face relations with him, especially with regards to the importance of familiarity in this process of communication. See David Goode, *A World without Words: The Social Construction of Children Born Deaf and Blind* (Philadelphia, 1994).

26. The disappearance of the role of assistants and instruments in the construction of scientific facts is related partly to the scientist’s rhetoric when he constructs himself as the sole author of facts produced and partly to scientific practice itself, which aims to objectify facts. Yet, one can add, that which is normally hidden in a single corporal envelope when it concerns a scientist endowed with an active body with Hawking becomes exteriorized because of his handicap. We can thus see, with him, that which is normally visible but intentionally hidden, as well as that which is normally present but constitutionally hidden.

27. After the present text was completed, I discovered a fascinating book written by Albert Robillard (*Meaning of a Disability: The Lived Experience of Paralysis* [Philadelphia, 1999]). Robillard has ALS and describes—with the tools of ethnomethodology—the implication of his disease for interactions in daily life. I found, from the point of view of one who can’t read an interaction, the same things he found through his difficulty with everyday interactions and from

Conversation analysts have shown that “the organization of taking turns to talk is fundamental to conversation, as well as to other speech-exchange systems.”<sup>28</sup> The turns that an interview system organizes are formalized as questions and answers. In this context, Hawking’s response to my first question occurs ten minutes after I asked it—and after my second question—as I was repeating it for the second time. And, instead of responding to my third question, he asked me a question instead. The normal order of question and answer seems inverted.<sup>29</sup> It is known that “discontinuities occur when, at some transition-relevance place, a current speaker has stopped, no speaker starts (or continues), and the ensuing space of nontalk constitutes itself as more than a gap—not a gap, but a lapse” (“SS,” p. 714). Hawking’s profound silence, when it occurs at transition-relevance places, becomes a lapse, which creates a discontinuity in the flow of the conversation.<sup>30</sup> Because of the organization of turn taking, “a participant, willing to speak next if selected to do so, will need to listen to each utterance and analyse it at least to find whether or not it selects him as next speaker” (“SS,” pp. 727–28). In this case, we are not in the relation of talk-listen-talk, but of talk-read-talk (plus sometimes listen), and one notices the oddity of reading and listening at the same time. Because the question “Why you?” written on his computer is accompanied not by his voice but by an artificial voice and my eyes are not trained to read this interaction, I won’t be able to know that it is addressed to me.

Moreover, one is struck (because of their absence) by the importance of the markers inscribed in the body, which enable us to open, sustain, and close the interaction.<sup>31</sup> For Merleau-Ponty

the body appears as a phenomenon with which we are familiar; we perceive human behavior through the body. . . . The body, having a

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his feelings of isolation: the role of mutual gaze, what he calls “real time” in conversation, and in general the use of the body and voice to create and maintain participation in a social setting.

28. Harvey Sacks, Emanuel A. Schegloff, and Gail Jefferson, “A Simplest Systematics for the Organization of Turn-Taking for Conversation,” *Language* 50 (Dec. 1974): 696–735; hereafter abbreviated “SS.”

29. The question “Why you?” can be interpreted as being a trick to give him time to respond to my question. This bridges the gap occurring during the conversation, just as his assistants do when, during conferences, they explain the workings of Hawking’s computer while he takes the time to compose his answers to questions. We can also say that in a certain sense he does not respond like a machine. Nevertheless, Hawking would not be able to pass the Turing Test; because of his slowness at answering questions, someone who is not in his presence might have the feeling that he or she is communicating with a machine rather than with a human being.

30. We see also different ways in which his silence is significant for pragmatic reasons. This silence encourages people to project and create meanings.

31. It is important to note that researchers who focus on gestural language reveal its unnoticed, because hidden, presence by “tricks of language.” In my case, its presence is noticed because it is absent. Formal language is thus totally detached from gestural language.

higher degree of integration by reason of its structuration as a *human* body, is the visible expression of meaningful behavior, communicable to other "ego-bodies" which are, likewise, centers of meaning and points of mutual encounters. This already implies an intersubjectivity . . . but actually it will be seen that the encounter is necessary for the very "realization" of the subject.<sup>32</sup>

Put another way, Erving Goffman has shown that what constitutes and makes face-to-face interaction possible is precisely the immediacy of the corporeal copresence, which implies using signs or clues displayed by bodies in order to regulate the interaction. With Hawking, all the markers normally inscribed in a body (gestures, glances, positions, intonation, silences, sighs, or throat clearing) are missing. Instead, I have a more or less approving smile on a statuary body and a discourse fixed on a machine, punctuated by signs relating either to physical needs (legs, suction); or to wishes, feelings, or thoughts (I hope, I think, I am); or questions ("Why you?"); or orders ("Get Tom!"). Finally, an important element of an interaction (especially for turn taking) is mutual perception, which is made impossible in this context because of the mediation of the computer; we don't look at each other, we both look instead at the computer.<sup>33</sup>

If we follow Goffman, gestural movements are signifying movements; they have a conventional meaning. A smile does not have the same meaning in Japan as it does in America. But, more than that, gestures reveal a reality that becomes visible only through their mediation, which is that of a person. Bodies, in other words, become surfaces on which the definitions of the self and the other are inscribed. This piling up of mutual self-definitions (and denunciation or ratification of these definitions in the course of the interaction) takes place between bodies. My first observation, then, is that this human body, through the mediation of the other body (the computer) talks only about itself and tells me nothing about myself. That is why I don't know what to do. I'm interacting not with *a* body but with an *extended body*.

In other words, I am confronted with:

1. An "*illegible*" *body*, for our eyes are not used to interpreting a body without movements, gestures, and a voice that betray something about their author and enable us to attribute intentions to it and to read in it whatever it incorporates and says about ourselves.
2. A *computer* that, by its presence, reconfigures the spatial arrangement necessary for the mutual perception making an interaction possible and

32. Mary Rose Barral, *Merleau-Ponty: The Role of the Body-Subject in Interpersonal Relations* (Pittsburgh, 1965), pp. 92–93.

33. See, for example, Adam Kendon, "Some Functions of Gaze Direction in Social Interaction," *Acta Psychologica* 26 (1967): 22–63.

that, by the waiting required by the task of writing, destroys the time scale necessary for the smooth running of a conversation. For example, answers do not directly follow the questions preceding them. Moreover, the artificial voice is next to Hawking, where it repeats a written discourse. By acting as the echo of the writing, the voice disturbs understanding. It repeats but does not talk. Finally, functions generally dissociated in a conversation, like listening and reading, are simultaneously combined.

3. *A discourse*, which is difficult to read. According to Paul Ricoeur:

Dialogue is an exchange of questions and answers; there is no exchange of this sort between the writer and the reader. The writer does not respond to the reader. Rather, the book divides the act of writing and the act of reading into two sides, between which there is no communication. The reader is absent from the act of writing; the writer is absent from the act of reading. The text thus produces a double eclipse of the reader and the writer. It thereby replaces the relation of dialogue, which directly connects the voice of one to the hearing of the other.<sup>34</sup>

Although written, this discourse does not have the same modalities as writing, such as a certain fluidity insofar as it is detachable from its conditions of production and the inclusion of an invisible and unknown reader; in this case, this text is intended for me, it speaks to me. Neither does it have the modalities of a conversation. In a spoken discourse “the subjective intention of the speaking subject and the meaning of the discourse overlap each other in such way that it is the same thing to understand what the speaker means and what his discourse means. . . . It is almost the same thing to ask ‘what do you mean?’ and ‘What does that mean?’”<sup>35</sup> But, in the case of Hawking, this equivalency is ruptured. The discourse does not come from the mouth of its author; his *I* is in front of him—on the screen—and the markers of turn taking are absent. Moreover, what this writing does fix is not the said of speaking but the event of speaking.<sup>36</sup> Similarly, this discourse fails to show the modalities of interaction by, for example, email, for the author is present—juxtaposed to his text—and at my side. I therefore have to read a text without having the punctuation of the author’s voice, yet with the silent author beside me.

What is normally attributed to the body, the machine, or the mind is inverted. The specificity of a machine (as opposed to a body) is that it cannot

34. Ricoeur, *Hermeneutics and the Human Sciences*, pp. 146–47.

35. *Ibid.*, p. 200.

36. See *ibid.*, p. 199. On the contrary, Ricoeur argues that what writing fixes is not the event of speaking but the said of speaking.





FIGURE 2. "The view from behind the desk." From Stephen Hawking's *"A Brief History of Time": A Reader's Companion*, ed. Stephen Hawking and Gene Stone (New York, 1992), p. 167.

repair itself; this is precisely what characterizes Hawking's body. The specificity of a mind is that it produces thoughts, language, desires; in Hawking's case they are legible and emanate from the machine. So where is his body? (See fig. 2.)

### Learning about Hawking's Extended Body

Moments of crisis shed the first light on the matter. When something goes wrong with Hawking's body (pain, discomfort) or with the computer (failure), the assistants—*ever-present* and ready to intervene, although discreet and relatively unobtrusive during the interview—are mobilized by the computer and the activation of its voice function. They try to be immediately operational by reading the words displayed on the computer screen or by reading Hawking's face or by touching the various apparatuses surrounding him, just as a doctor would feel the various parts of a patient's body to detect painful areas. Hawking is at the same time the surgeon in command of the bodies of others and the patient being taken care of by the others.<sup>37</sup> Thus the parts of the human or mechanical body to be repaired are identified (*legs* or *chair-in*) and/or the operations to be carried out are

37. See Stefan Hirschauer, "The Manufacture of Bodies in Surgery," *Social Studies of Science* 21 (May 1991): 279–319.

indicated (*suction* or *get Tom*). In this process Hawking's voice acts as an indicator: *yes*, *no*, and so on. The distribution of tasks seems fairly well codified, though it can also be negotiated during the action. When the voice says *legs* or *suction*, there is no ambiguity. The nurse is immediately operational, whereas *chair-in* causes a moment's hesitation. The nurse wants to extend her competence to repair the computer but, as we have seen, Hawking does not want her to intervene. That is Tom's domain. The artificial voice is important because it is used to alert this small collective of assistants when there is a problem. It thus enables them to have a relatively large degree of autonomy. They are not forced to be constantly breathing down Hawking's neck. The nurse, alerted by the voice, can serve as a relay between Hawking and Tom. It is also through this voice that a semblance of normal conversation with me can be created (*yes*, *no*, *Why you?*). He does not, however, use it to provide formal answers to my questions or to communicate with his assistants when they are around him. Thus, communication with Hawking will involve the reading of signs on the computer and/or signs expressed in body language (the movement of his eyebrows or mouth)—skills acquired over time by the people working with him. Between body language and formal writing there is a gap: that of speaking. We also notice a strange inversion: a deaf person would read the lips of the person who speaks, the assistant reads Hawking's face. As Tom put it:

T.A.: He doesn't need to use his voice, really, unless I'm facing him and on the table, he talks with his face, when I am talking about work or whatever, or doing slides for him, *there's a lot putting words in Stephen's mouth*, and he says that's a yes and that's a no, there's a lot of that . . . so yeah, he talks with his face . . .

H.M.: So, I imagine that you anticipate a lot!

T.A.: Yes, there aren't many answers. . . . Like if I say: Do you want this published? I'll see how he reacts. Or if he's talking about a meeting, and I go: tomorrow? . . . It's pretty obvious he either wants it tomorrow or after he gets back. . . . You don't have to wait for him to say it, and if there's a time, I'll suggest a time, and I'll suggest other times. . . . But yes, you put words in his mouth, and he communicates with his face a lot, so there's a lot of that, *he doesn't need to use his voice, as you look on the screen*. . . . You will see that, surely.

We can see a strange symmetry between, on the one hand, a choice of responses proposed by either the assistants or the machine and Hawking responding by *yes* or *no* and, on the other, the binary logic of a computer (*true/false*, *yes/no*) that proposes the choice of sentences and words. One can notice also an extraordinary similarity between the interaction between Hawking

and his technicians and Hawking and his computer; both seem to have been trained or programmed to operate in the same way. Each gives Hawking a series of options to choose from, after which the technical object, technician, nurse, or computer performs it. Hawking's assistant's operation duplicates the computer's except that rather than responding to the subtle movement of Hawking's fingers, he responds to the subtle movement of his eyes.<sup>38</sup>

Hawking's special conditions impose the mechanization (the hierarchization, standardization, and routinization) of his human/machine based environment.<sup>39</sup> What is at stake behind this coordination and hierarchization of tasks is human life. A false maneuver in any of these operations can result in death.<sup>40</sup> The graduate assistant will characterize his job not only by its high level of responsibility (in the end what is at stake is the life of a human being) but also by the availability required ("we are on call twenty-four hours a day") and the difficulty of living constantly together. What is also at stake is Hawking's ability to work and to maintain his status as Stephen Hawking the Genius Physicist. This network of competences allows a yes or a no to be operational. A yes answer to the question, Do you want to go to this conference? will enable him to travel from one end of the earth to the other without more than twitching an eyebrow.<sup>41</sup>

38. Hawking's role can be compared to the role of the psychoanalyst in general. I am also thinking about Garfinkel's discussion of an experimenter, falsely represented as a student counselor in training, who responds to his patients' questions in a predetermined sequence of yes and no answers without taking into account the content of the patients' questions. The patients do almost all the work of the counselor (like the people around Hawking, his students, colleagues, and assistants) but still attribute all the merit to the counselor himself. See Garfinkel, *Studies in Ethnomethodology*, pp. 76–103. On the role of fulfillment in interactions between persons or between persons and machines, see Harry Collins, *Artificial Experts: Social Knowledge and Intelligent Machines* (Cambridge, Mass., 1990); Collins, "The Turing Test and Language Skills," in *Technology in Working Order*, ed. Graham Button (London, 1993), pp. 231–45; Lucy Suchman, *Plans and Situated Action: The Problem of Human-Machine Communication* (Cambridge, 1987); Donald MacKenzie, *Knowing Machines: Essays on Technical Change* (Cambridge, Mass., 1996), pp. 131–59; Joseph Weizenbaum, "ELIZA: A Computer Program for the Study of Natural Language Communication between Man and Machine," *Communications of the ACM* 1 (1966): 36–45; and Sherry Turkle, *Life on the Screen: Identity in the Age of the Internet* (New York, 1997), pp. 102–24.

39. On the fundamental role of routine as an intersubjectively shared resource in negotiating daily life, see Goode, *A World without Words*, pp. 65–73. In a certain sense, we can say that we all have a more or less routinized environment around us. Nevertheless, the able-bodied seem able to navigate more or less freely in this environment. Hawking never manages to escape from this little microenvironment, this extended body that I have just described.

40. In this sense, there are obvious parallels between Hirschauer's article on surgery (Hirschauer, "The Manufacture of Bodies in Surgery"), Edwin Hutchins on the functioning of a cockpit (Edwin Hutchins, "How a Cockpit Remembers Its Speeds," *Cognitive Science* 19 [July–Sept. 1985]: 265–88), and this article on Hawking. In a forthcoming article I also describe how this extended body travels from one point to another and the complications this implies.

41. In *A World without Words*, Goode considers the way in which children who are totally deprived of formal language manage to communicate with their bodies when they are caught in a network of competences organized around them and with them. I find myself confronted with the

### Hawking's Three Bodies

According to Polanyi:

To use language in speech, reading and writing, is to extend our bodily equipment and become intelligent human beings. We may say that when we learn to use language, or a probe, or a tool, and thus make ourselves aware of these things as we are of our body, we *interiorize* these things and *make ourselves dwell in them*.<sup>42</sup>

The computer, in the present context, has thus become an appendix of the human body owing to the acquisition of a number of skills that Hawking has mastered: the rapidity with which he touches his commutator, the manipulation of certain operations for selecting words in the construction of sentences, the choice of the right keys to activate the synthetic voice.<sup>43</sup> As one would say of an individual who has mastered all the subtleties of a new language, he has, as one of his students points out, "become totally 'fluent' with his computer." The designer of Hawking's software, Walter Woltosz, recalls that Hawking could be much faster with using a new software called EZ Keys at carrying on a conversation, but Hawking thinks that he is faster on Equalizer: "It is burned into his brain. He has been using it for fifteen years . . . fifteen hours a day and he knows it so well that he is very comfortable with it. . . . It really is an extension of himself now."<sup>44</sup> As Polanyi says, "Every time we assimilate a tool to our body our identity undergoes some change; our person expands into new modes of being,"<sup>45</sup> or, as Drew Leder suggests, in the tradition of Merleau-Ponty: "The true relation between body and instrument, phenomenological rather than crudely materialistic in character, is only revealed when we reverse the analogy. It is not that the body is like a tool, but that the tool is like a second sort of body, incorporated into and extending our corporeal powers."<sup>46</sup> "To incorporate a tool is to redesign one's extended body until its extremities expressly mesh

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opposite situation, for Hawking is endowed with formal language but is deprived of a voice and partly of the gestural language to express it (except if you are familiar with it). It is nevertheless interesting to note how a yes or no from him takes on a different meaning in the various networks of competences in which it is set. In this sense, we might wonder whether for Hawking language is a property of this network or of the individual.

42. Michael Polanyi, *Knowing and Being*, ed. Marjorie Grene (London, 1969), p. 148.

43. "A highly skilled user like Hawking can use the software to speak about 20 words per minute in conversations, compared with 150 to 200 for natural speakers" (David Colker, "Giving a Voice to the Voiceless; Technology: Software Enables Severely Disabled People to Communicate by the Touch of a Button," *Los Angeles Times*, 13 May 1997, p. 8).

44. Woltosz, interview, Dec. 2000.

45. Polanyi, *The Study of Man: The Lindsay Memorial Lectures* (Chicago, 1959), p. 31; hereafter abbreviated SM.

46. Drew Leder, *The Absent Body* (Chicago, 1990), p. 179 n. 70; hereafter abbreviated AB.

with the world" (AB, p. 34).<sup>47</sup> In this sense, this tool, the extension of his human body, enables Hawking to mobilize and extend his competences by affording him the ability to select information, read and write scientific articles, prepare talks, and communicate with colleagues informally or by email conferences.<sup>48</sup> As he himself put it in a recent interview conducted via email: "Without this software or something like it, I would have been cut off and unable to carry on as a physicist."<sup>49</sup>

However, our analogy with the world of phenomenology stops here. Indeed, for Polanyi the body is, above all, *the* main instrument through which the world is apprehended: "Our body is always in use as the basic instrument of our intellectual and practical control over our surroundings" (SM, p. 31). "Our body is the ultimate instrument of all our external knowledge, whether intellectual or practical."<sup>50</sup> With Hawking we have seen that it is not his flesh-and-blood body that becomes "the basic instrument of [his] intellectual and practical control over [his] surroundings" but the computer itself. The computer is the center of the network through which (almost) all interactions are negotiated because they are rendered visible and public (the computer distributes competences between what is appropriate to/for Hawking's flesh-and-blood body, his identity, his assistants, his computer, and me). The computer makes the interaction possible, between Hawking and myself (Hawking can write, answer my questions, and create a semblance of conversation), between Hawking and his assistants (he can mobilize them or point out problems that interrupt the smooth functioning of this interaction), and between Hawking and himself (it allows him to write his flesh-and-blood body). By allowing the mobilization of assistants, it allows the maintenance of the flesh-and-blood body and of the machine and, ultimately, the writing of a text.

Hawking can move only one finger and a few muscles of his face. The structure of the system (the way in which language appears and is organized in and on his computer, such as the alphabetical organization of words in columns, the gathering of a vocabulary most commonly used, possible abbreviations, word predictions, instant phrases, and his artificial voice) and

47. See Polanyi, *Knowing and Being*, p. 145 and M. Merleau-Ponty, *Phenomenology of Perception*, trans. Colin Smith (London, 1962), p. 143, or the example of the handheld tool discussed by Martin Heidegger, *Being and Time*, trans. John Macquarrie and Edward Robinson (New York, 1962), pp. 95–107. Don Ihde describes a similar phenomenon: "I may describe these relations as *embodiment relations*, relations in which the machine displays some kind of partial transparency so that it itself does not become objectified or thematic, but is taken into my experiencing of what is other in the World" (quoted in AB, p. 179 n. 69).

48. Hawking can use EZ Keys as an interface with Equalizer when he wants to do email or word processing or use the internet.

49. Quoted in Colker, "Giving a Voice to the Voiceless," p. 8.

50. Polanyi, *The Tacit Dimension* (Garden City, N.Y., 1966), p. 15.

everything it enables him to do construct his intellectual competences as much as they extend them. In other words, these instruments are not merely transparent extensions through which his intelligence is diffused, for they comprise it as much as they allow it to be diffused.

The body is the privileged instrument or, as Polanyi but also Merleau-Ponty says, trying again to muddle the dichotomy between instruments and the body, instruments are also another part of the body. With Hawking, the body is not the privileged instrument, and the instruments are much more than another part of the body; they make the body present as much as they comprise it. As we have seen, this system enables him to mediate his relationship with his own human body by excorporating or making visible the normally invisible functioning or dysfunctioning of certain neuromotor operations. Owing to the computer's mediation, Hawking can immediately allow the pain in his human body to be located, by writing *legs*, for example, on his computer. By activating the synthetic voice, he can mobilize the competences of his nurse so that she is immediately operational to remedy this dysfunctioning. The human body is thus taken care of by the nurse, through the computer, which serves to alert her and to pinpoint the problem. In this way, through these different mediations, the functioning of this human body becomes visible (for us as observers and for the nurse), externalized and collectivized to such an extent that, as long as the body is taken care of by the relay of machines, men, and women, Hawking no longer thinks about it (like when we swallow without thinking about it).<sup>51</sup> This sequence consists of a multitude of actions: feeling of pain in the human body, *legs* displayed on the computer, *legs* spoken by the computer, mobilization of the nurse, movement of Hawking's leg, activation of his nurse's arm, leg rested—all of which have become the movement of one and the same self-repairing body. For Hawking, the movement of this natural, collectivized body has become invisible, as a body endowed with its own ability to move would be.<sup>52</sup>

51. It would be interesting to explore the relationship between memory and automatism. This echoes what David, who is quadriplegic, said to me during an interview conducted in December 2000: "I don't experience myself as a disabled individual all the time. I am aware of it but it is not the focus of my life. At some point in time disability stopped being the main thing in my life and started just being a major inconvenience, which pops its head up every once in a while, especially when I need something that I can't get."

52. This point agrees with what Leder says:

I do not notice my body, but neither do I, for the most part, notice the bed on which I sleep, the clothes I wear, the chair on which I sit down to breakfast, the car I drive to work. I live in bodies beyond bodies, clothes, furniture, room, house, city, recapitulating in ever expanding circles aspects of my corporeality. As such, it is not simply my surface organs that disappear but entire regions of the world with which I dwell in intimacy. [AB, p. 35]

On the other hand, the computer also makes it possible to visualize all dysfunctioning in the mechanical body (the computer is reflexive or says something about itself), for example, *chair-in*. Yet we have seen that dysfunctioning in Hawking's human body does not preclude the activity of writing, whereas dysfunctioning in the mechanical body precludes all interaction. It therefore also precludes the possibility of mobilizing the assistants and allowing the human body or the mechanical body to be repaired, as well as all possibility of writing.<sup>53</sup> As I observe Hawking orchestrating the functioning of his extended body through the mediation of the computer, I cannot help but think of the materialization of the "pilot in his vessel," a metaphor used by Descartes to describe the difficult union of body and soul:

Nature likewise teaches me by these sensations of pain, hunger, thirst, etc., that I am not only lodged in my body as a pilot in a vessel, but that I am besides so intimately conjoined, and as it were intermixed with it, that my mind and body compose a certain unity. For if this were not the case, I should not feel pain when my body is hurt, seeing I am merely a thinking thing, but should perceive the wound by the understanding alone, just as a pilot perceives by sight when any part of his vessel is damaged; and when my body has need of food or drink, I should have a clear knowledge of this, and not be made aware of it by the confused sensations of hunger and thirst: for, in truth, all these sensations of hunger, thirst, pain, etc., are nothing more than certain confused modes of thinking, arising from the union and apparent fusion of mind and body.<sup>54</sup>

Whereas Hawking seems to feel pain in his human body, the nurse and Auld (indispensable elements of his extended body) give us the impression that they perceive as he does, through the intermediary of the computer

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This transparency is never complete; as Ihde points out there is always an "echo focus," a subliminal awareness of the instrument at my body boundary. Moreover, especially at times of malfunction, the tool can become thematically central. [AB, p. 179 n. 69]

When incorporating a tool, one's concurrent incorporation by it may have a series of unanticipated results. As Ihde notes, every instrument imposes an "amplification-reduction" structure on one's natural capacities. The telephone, while allowing communication through its amplification of voice, in other ways yields a reduced encounter devoid of direct sight and touch. As a result of such transformations, each technology provides a "telic inclination" toward certain modified modes of interaction. [AB, p. 181 n. 72]

See also a similar argument in Don Ihde, *Bodies in Technology* (Minneapolis, 2002), chap. 1. Hence, the problem this implies in my mode of communication with Hawking.

54. René Descartes, *The "Meditations" and Selections from the "Principles,"* trans. John Veitch (Chicago, 1913), p. 94.

(another part of the extended body), “the wound by the understanding alone, just as a pilot perceives by sight when any part of his vessel is damaged.”

By unfolding itself and making itself visible through the machine, Hawking's flesh-and-blood body calls into question that which constitutes the primacy of the body itself, distinguishing it definitively from other objects. The body can't be “a mere object of which he is conscious as he is of every other ‘thing’ in the world.” If so, he would have “embraced the position of the ‘pure subject’ who is a *spectator* before the world and also before his own body which he considers as part of the world of objects.”<sup>55</sup> Because “in so far as it sees or touches the world, my body can therefore be neither seen nor touched. What prevents its ever being an object, ever being ‘completely constituted’ is that it is that by which there are objects. It is neither tangible nor visible in so far as it is that which sees and touches.”<sup>56</sup> Indeed, “I am not in front of my body, I am in it, or rather I am it.”<sup>57</sup> In other words, the perceiving body is always behind the perceived object, which is why my body “offers itself obstinately ‘from the same side’ without my being able to go around it.”<sup>58</sup> In a way we can say that Hawking's flesh-and-blood body is registering in the same text as his speech; they both appear on the screen in front of him. Without this work of objectification of the body, Hawking cannot be his body. The body is felt but also represented as an exterior thing before it can do anything. With him the distinction between body-object and body-subject disappears. The human body conceived as a body-object, that is, which “allows between its parts or between itself and other objects only external or mechanical relations” has become his own body or the body-subject, that is, “his’ body as the center of ‘his’ existence, as the power both to act and to perceive and as a means for the subject to be part of the world.”<sup>59</sup> We thus see the movement of a certain form of absence-presence that characterizes the body as Leder described it, but in another form. For Leder, the body is always present but situated in the background. Firstly, this is because our insides and even certain visible parts of our exterior, such as the area around our eyes, are hidden. Secondly, the possibility of allowing

55. Barral, *Merleau-Ponty*, p. 94. Also, according to Polanyi, “We must realize then also that our own body has a special place in the universe: we never attend to our body as an object in itself” (*SM*, p. 31). Or, “Dwelling in our body clearly enables us to attend *from* it to things outside, while an external observer will tend to look *at* things happening in the body, seeing it as an object or as a machine” (Polanyi, *Knowing and Being*, p. 148).

56. Merleau-Ponty, *Phenomenology of Perception*, p. 92.

57. *Ibid.*, p. 150.

58. Merleau-Ponty, *The Structure of Behavior*, trans. Alden L. Fisher (Boston, 1963), p. 213.

59. Louis-Marie Morfaux, *Vocabulaire de la philosophie et des sciences humaines* (Paris, 1980), p. 67, s.v. *corps*; translation mine.



ourselves to be immersed in a situation is related to a form of forgetting ourselves. In situations of pain and discomfort the body becomes present again. It is subjected to a form of dys-appearance or absence of absence. With Hawking, in moments of discomfort, the body becomes present again, but, more than that, it sets itself before him and before those taking care of him/it. It is thus collectivized, taken charge of, while the collective of humans and nonhumans becomes invisible again and disappears behind him; it has become his body. All his activity is then directed towards the activity of writing a text.

Thus, by incorporating the computer as the physical extension of his own body, Hawking can mobilize and extend his intellectual capabilities. He can excorporate and make visible the neuromotor functions of his body,<sup>60</sup> as well as all the dysfunctions in the technical system surrounding him, and can mobilize the human competences required for the smooth functioning of this collective. Finally, and consequently, he can simultaneously fulfill his role as Lucasian Professor of Mathematics. That is to say, he can write his own history by answering my questions on his role in the scientific community, the intellectual authority of his status, the changes it underwent over time, the characteristics he would like to see in his successors, and the importance of his peers' role as regards the authority of his own status. We thus witness the operation, maintenance, and (*non*)*simultaneous* articulation, through the mediation of the computer, of what I call, to paraphrase Ernst Kantorowicz, Hawking's three bodies: first, a *natural human body* collectivized—this natural human body cannot function without this collective; second, a *collective body* (human body/computer/assistants) naturalized—this collective body becomes "his" body; and third, a *sacred body*—the representative of the Lucasian Chair of Mathematics (which "is not subject to Passions as the other is, nor to Death, for as to this Body the [Professor] never dies)."<sup>61</sup>

However, all the interactions described above, relating to the functioning of this extended body, will totally disappear in the official text handed to me. Even though, as I tried to show previously,<sup>62</sup> Hawking attaches his sin-

60. He makes visible the dysfunctions of his flesh-and-blood body but also all his normal needs like the need to eat and drink.

61. Quoted in Ernst H. Kantorowicz, *The King's Two Bodies: A Study in Medieval Political Theology* (Princeton, N.J., 1957), p. 13. Kantorowicz adds, "Interesting, however, is the fact that this 'incarnation' of the body politic in a king of flesh not only does away with the human imperfections of the body natural, but conveys 'immortality' to the individual king as King, that is, with regard to his superbodily," (*ibid.*, p. 13). We also witness a dissociation between corporeal, mental, and social phenomena, such as the construction of the self, "normally" inscribed in or displayed *simultaneously* by "one" body.

62. See Miallet, "Do Angels Have Bodies?"

gular and mortal being to the Corporation of Scientific Geniuses by using his mortal body to place himself in the tradition of his peers,<sup>63</sup> to establish the paternity of his ideas,<sup>64</sup> or to justify the choice of his fields of research,<sup>65</sup> he also effaces all physical evidence of his mortal body, despite its indispensability in the production of this paper. Indeed, Hawking is able to switch between the *talk* and *write* functions of the software. When the program is set to *write*, what he composes is stored in a file as in a word processor, but when set on *talk*, things are just written, said, and then cleared. Thus, sleeping like the unconscious, the computer's memory is an immense repository of formal written articles, lectures, and interviews, while all the words concerning informal interactions evaporate, depending on whether or not Hawking deems them worthy of being saved. Accordingly, owing to the computer, I could "see" the (non)simultaneous functioning of these three bodies; owing to the same computer (assuming for example that Hawking had sent me his history by post), we might not have been able to "see" how the dissociation between these three bodies takes place. That is, how Hawking, through the mediation of the machine, could both make his extended body function and, as he responded to my questions, construct his identity as Stephen Hawking, Lucasian Professor of Mathematics, while at the same time, as the interview moved towards its completion, effacing all presence of his physical body, the people who surround him, and even me. In his case, the noise and the conditions that made the interview possible were filtered out, not (just) afterwards by the analyst, as it is usually done, but during the interview itself. By the end of the interview, Hawking's responses—which appeared to me on his screen—were printed out as *sui generis* statements without any trace of their original conditions of production, while on my tape recorder only the sound of my own voice remained—questions asked to the invisible and echoed by silence. To paraphrase Barthes, what this machine of genius was supposed to produce was not only equations but also a (self-effacing) man.<sup>66</sup>

63. He likes to emphasize that he was born exactly 300 years after the death of Galileo, or that the last visible trace of his "acting" body is appended in the form of a signature in the "golden book" that immortalized him and introduced him into the pantheon of the great in 1979 when he was named Lucasian Chair of Mathematics, a post formerly occupied by Newton.

64. "One evening . . . shortly after the birth of my daughter, Lucy, I started to think about black holes as I was getting into bed. My disability makes this a rather slow process, so I had plenty of time" (Hawking, *A Brief History of Time* [London, 1988], p. 99). After the publication of *A Brief History of Time* in 1988, this became the established version of his discovery.

65. He emphasized that, although he had the misfortune of having this motor neuron disease, he had been lucky in every other respect and notably in his choice of theoretical physics because everything is in the mind, which is why his infirmity had not been a serious handicap.

66. The text handed to me is doubly disembodied. On one hand, its conditions of production have been erased, but this aporia demands that the text, if it is to be coherent, refers back to this

## Epilogue

We have thus followed a strange movement, a sort of coming and going between perceiving subjects and perceived objects, a zoom effect like that of a camera trying to set the right distance with its object in order to achieve a maximum degree of clarity—total transparency. On the one hand, we have seen how the tricky adjustment between the multiplicity of stabilized representations of Hawking and my face-to-face interaction with him has fragmented his identity. Like an impressionist painting, distance allows us to see what proximity causes us to lose. The layering of the multiplicity of narratives about Hawking reveals the stable identity of an individual that face-to-face contact does not allow. The closer we get to the object-subject, the further we move from it. We no longer know who or where he is. All the categories we normally use in thinking about a person, a body, a machine, a mind, an interaction, a conversation, a text, and speech are blurred. In his presence, Hawking seems even more difficult to seize than we had imagined. The reason is, first, because, as we have shown, the depth of the social body (the one that talks of the other as much as of itself and allows the accumulation of self-definitions), normally visible but unnoticed in any interaction, is absent in his case. Second, the internal functioning of the flesh-and-blood body, normally invisible, becomes visible and distributed here (the *I* and *legs* are mixed in the same discourse and are detached from the body; they are before him while his voice is next to him, and the flesh-and-blood body is collectively taken care of) and omnipresent (it appears at different moments during the interaction). Third, the machine, by the coordination of the actions it allows, disturbs the interaction as much as it creates it. This is where the confusion and fragmentation of identity is pursued by recomposition. Here we follow, step by step, how the machine to which this silent body is attached makes the components of a collective body visible and mobilizes a part of it—the assistants—to repair both the flesh-and-blood body and the mechanical body. We now see how this collective body becomes the man's own body (his extended body), while at the same time making possible the construction of his sacred—corporate—body as Lucasian Professor. Lastly, we are able to follow how the machine—another part of this collective body—gradually erases the presence of Hawking's flesh-and-blood body, his extended body, and my own body, all of which are conditions, in this interaction, for the construction of his sacred body. After having made visible what is normally invisible in an interview (the

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absence—that is, to the conditions that made it possible. On the other hand, as it is about Stephen Hawking, the Lucasian Professor of Mathematics who falls into the long lineage of his ancestors, his peculiar individual conditions are no longer significant as long as he produces scientific theories.

internal body), the machine makes invisible what is normally visible in the after-interview: its conditions of production. Thus, on the one hand, the machine disturbs the interaction as much as it creates it, while, on the other hand, the machine creates the interaction as much as it disassembles it, leaving that which will become public—a text that will be handed to me and stored in a part of the computer, waiting to be added to other narratives about Hawking and effacing that which has become private, the context of interaction. Has Hawking eluded us again?